MKALAMA DISTRICT COUNCIL

FORM TWO MOCK EXAMINATION 2025

PHYSICS MARKING SCHEME

1

| i | ii | iii | iv | v | vi | vii | viii | ix | x |
|---|---|---|---|---|---|---|---|---|---|
| D | A | C | A | D | C | B | C | A | C |

**@1Mark =10marks**

2

| i. | ii. | iii. | iv. | v. |
|---|---|---|---|---|
| C | E | G | A | F |

**@1Mark =5marks**

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3. **a**.

i.Do not overheat magnets

ii.Do not store or keep magnets near strong magnetic or electric fields. **@2Marks =6marks**

iii.Store magnets in pairs with the unlike poles facing each other.

**b.**

i.Magnetic recording media

ii.Credit debit and ATM cards **@1Mark =4marks**

iii.Common television and computer monitors

iv.Speakers and Microphones

4. **a.**

| Magnetization | Demagnetization |
|---|---|
| Process of making magnet | Process of destroying the magnetism in a magnetized material. |
| **4marks** | |

**b.**

**i.**If a permanent magnetism is heated or vibrated in the absence of an external field. **6marks**

ii. Randomly stroking one magnet with another can magnetise the magnet being stroked

iii.Repeated harmmering or dropping down the magnet.

5. **a**.

| capacitor | Capacitance |
|---|---|
| Device used to store electric charge | Ability of a capacitor to store electric charge |

**4marks**

b.

i.Paper or plastic capacitors

ii.Mica capacitor **3marks**

iii.Elecrolytic capacitor

c. “Like charge repel, unlike charge attract.” **3marks**

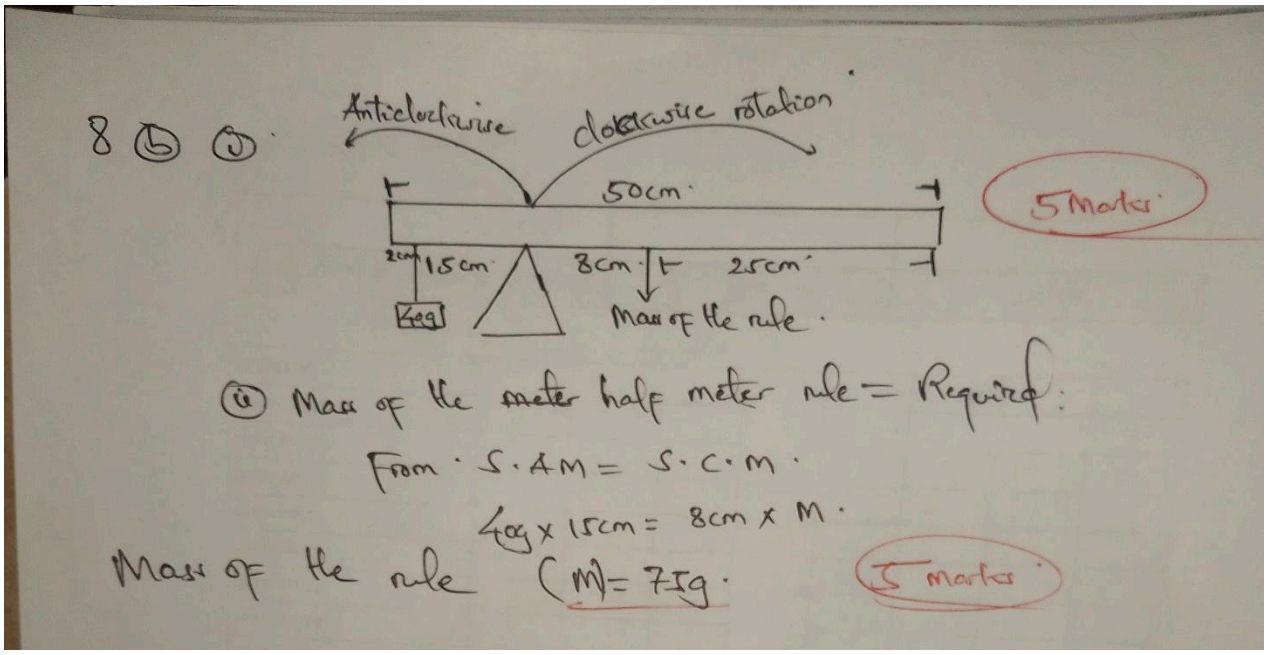
6. **a. Data given**
capacitance=0.5F
p.d= 18volts
amount of charge = required
from Amount of charge = CV
=0.5F x 18V
= 9C **5marks**

b **Data given**
Capacitance of capacitor $C_1$ = 5μF
Capacitance of capacitor $C_2$ = 10μF
Capacitance of capacitor $C_3$ = 15μF
From $1/C_T = 1/C_{1+} 1/C_{2+} 1/C_3$
$1/C_T$ = 1/5 + 1/10 +1/15
$1/C_T$ =0.667
$C_T$ = 1.499 μF
Value of a single capacitor to replace is 1.499 μF **5marks**

7. **a.**
**Data given.**
VR = 5
Efficiency = 80%
Load = 200N
Effort = Required
From Efficiency =MA/VR X 100%
80 = MA/5 X 100
MA = 4
From MA = Load/Effort
4 = 200N/Effort
**Effort = 50N** **6marks**

**b.**Because some of effort applied is used to overcome frictional forces in the moving parts of the system. 4**marks**

8. **a. a long spanner produce large moment of force than short spanner**



9. **a.**

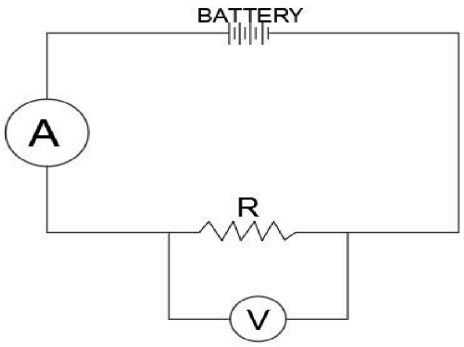
Data given
Load = 400N
Effort = 100N
MA = Required
MA = Load/Effort
400N/100/N
**MA = 4.** **4marks**

**b**. Data given
Distance moved by Effort = 10m
Distance moved by Load = 2m
MA= 4
VR= required
Efficiency= required
From VR = Distance moved by/Effort /Distance moved by Load
VR = 10m/2m
**VR = 5** **3marks**
Efficiency = MA/VR X 100%
= 4/5 X 100%
**Efficiency = 80%** 3**marks**

10. **a.**

Ammeter
Voltmeter
Connecting wire **5marks**
Dry cell
Resistor

b.




**5marks**

c. .Ammeter is connected in series with the resistor to measure the current flowing through the resistor

Voltmeter is connected in parallel across the resistor to measure potential difference across it **5marks**